

X: 128 frames $\times$ 1024 I/Q
 $\approx 2.6 \times 10^5$ real values

Stage 1: LSE
CPC encoder, InfoNCE
centered dilated conv

$\mathbf{z}_t$
128-dim

Stage 2: Diffusion
conditional reverse process
 $T_{\text{pred}} = 32$ frames

$\hat{\mathbf{Y}}$
 32×1024

Stage 3: T2F
1D CNN, 17M parameters
pretrained on ground truth

occupancy: 8 sub-blocks $\times$ 10 MHz
horizon $\approx 410 \mu\text{s}$ at 80 MHz